A FINITE MODULE OVER A NOETHERIAN DOMAIN IS REFLEXIVE EXACTLY WHEN IT EMBEDS INTO A FINITE FREE MODULE WITH TORSION FREE COKERNEL

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Allowed standard prerequisites (editorial).

- Reflexive module and the double dual map. definition at source lines 5148-5157 of the same chapter and the discussion following it. Supplies the notion appearing in condition (1).
- The kernel and cokernel of the natural map are torsion for finite modules, and a sequence with free and torsion free terms gives reflexivity. the author cites Lemma lemma-sequence-reflexive at proof line 5321. Supplies the implication from condition (2) to condition (1).
- Dual of a finite module and the exactness of Hom duality for a presentation by free modules. used at proof lines 5323-5330. Produces the exact sequence of the second condition from a presentation of the dual.

Source reference. Original lemma 0AV2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13670–13670. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 0EB8: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 17913–17913. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let R be a Noetherian domain. Let M be a finite R-module. The following are equivalent*

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(1) *M is reflexive,*
(2) *there exists a short exact sequence $0 \to M \to F \to N \to 0$ with F finite free and N torsion free.*

Proof. Observe that a finite free module is reflexive. By Lemma 0EB8 we see that (2) implies (1). Assume M is reflexive. Choose a presentation $R^{\oplus m} \to R^{\oplus n} \to \mathrm{Hom}_R(M, R) \to 0$. Dualizing we get an exact sequence

$$0 \to \mathrm{Hom}_R(\mathrm{Hom}_R(M, R), R) \to R^{\oplus n} \to N \to 0$$

with $N = \mathrm{Im}(R^{\oplus n} \to R^{\oplus m})$ a torsion free module. As $M = \mathrm{Hom}_R(\mathrm{Hom}_R(M, R), R)$ we get an exact sequence as in (2). □